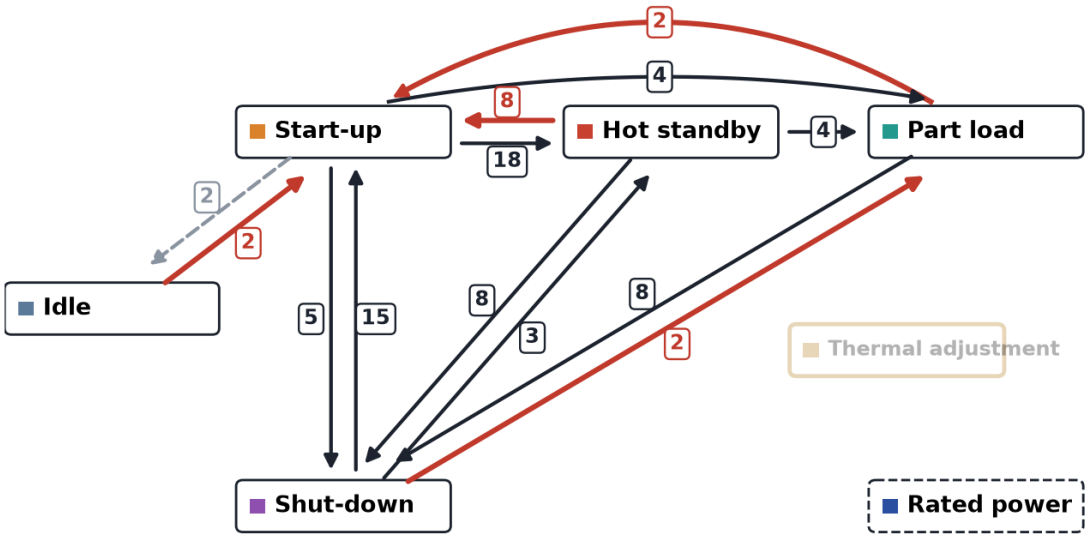
Stage Map Fix

SOFC stage labels, 39 days. Decided 2026-10-14. Pages: 1 The fix, 2 The new names, 3 What is still open.

The four impossible stage changes are gone. They were all a hot stack warming up or cooling down with no load, and they now have their own stage: **Thermal adjustment**.

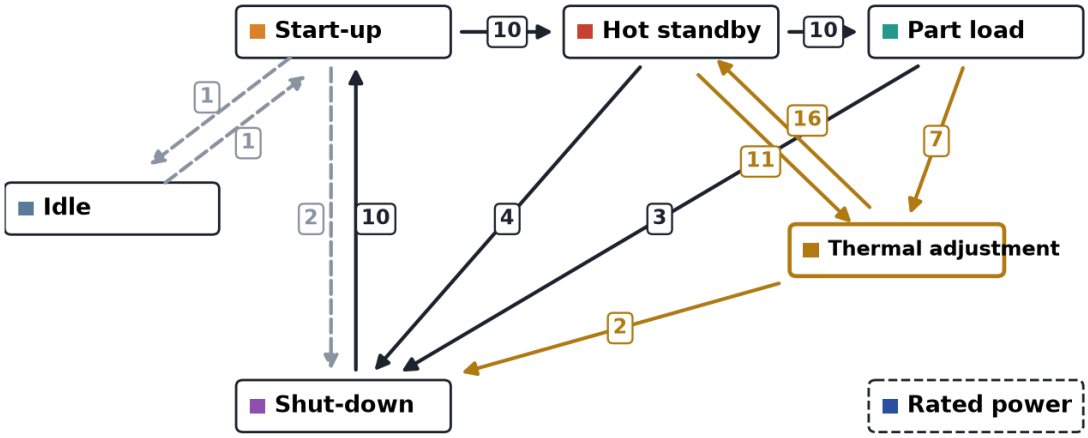
Why it matters: the synthetic-data generator learns what can follow what from these arrows. With the old labels it would have learned that a running plant can jump back into Start-up, which the real plant cannot do.

Before



Red arrows cannot happen on the real plant. Numbers = times seen in the 39 days.

After



Start-up only after Shut-down (plus once from Idle in the first file); Part load only after Hot standby; re-heats and cool-backs go through Thermal adjustment (gold). Grey dashed = seen, not in the design.

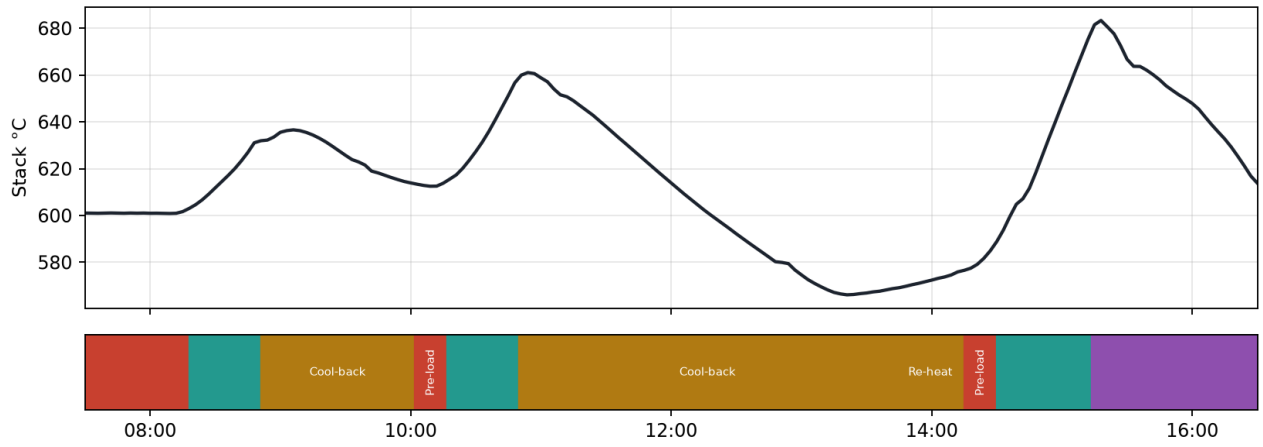
Arrow you flagged	Times	What the data shows	Now labelled

Hot standby → Start-up	8	Hot stack (510–610 °C) warming with no load: 3 climbs from the ~510 °C hold, 5 top-ups of 2–30 °C.	Thermal adjustment, re-heat
Part load → Start-up	2	Load stopped after 8–17 min (03-13, 03-19); hot stack kept warming.	Thermal adjustment, re-heat
Shut-down → Part load	2	Short pause between load runs (03-24, 03-25), about 630 → 615 °C.	Thermal adjustment, cool-back
Idle → Start-up	2	03-17/18: heat-up stopped at 190 °C, plant off 11 h, then continued. 10-30: first file.	03-17/18 one Start-up; 10-30 kept Idle

If you remember one thing: a hot stack changing temperature with no load is now its own stage, Thermal adjustment, so Start-up only comes after a Shut-down and Part load only after Hot standby.

The New Names

Name	What it means	Size
Thermal adjustment (new stage)	Hot stack, no power, temperature being moved. Re-heat (up) or Cool-back (down, then levels off or load starts). Was Start-up re-heat / Shut-down cool-back.	18 episodes, 45 h
Hot standby, Pre-load	Last 15 min before load starts, so Part load always follows Hot standby.	6 times, 1.6 h
Start-up, Heat-up interrupted	A heat-up stopped, plant off, then continued = one Start-up. 2026-03-17 20:08 → 03-18 07:11.	11 h
Shut-down, Cooling	Load off, then air or natural cooling while the stack is still warm.	308 h
Shut-down, At ambient (plant off)	Stack within 30 K of the room, waiting for the next Start-up.	78 h



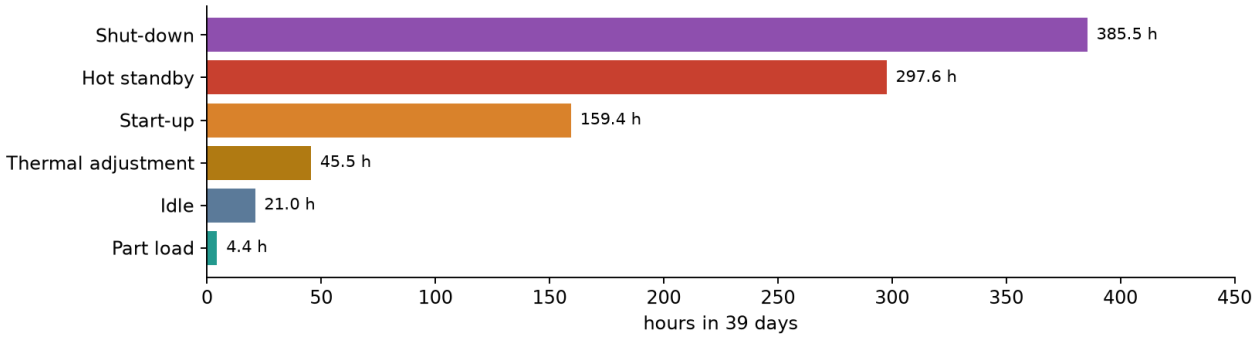
2026-03-25, 08:00–16:30. Before each of the three load runs (teal) there is now Hot standby (red, Pre-load). The pauses between runs are Thermal adjustment (gold). The day ends in a real Shut-down (purple, Cooling).

If you remember one thing: Thermal adjustment is the hot, no-power in-between stage; Shut-down now says whether the stack is still cooling or already at room temperature.

Still Open

The four arrows you flagged are gone. Three smaller arrows outside the design remain, plus one questionable re-heat.

Remaining arrow / flag	Times	What it is	Your call
Idle → Start-up	1	2025-10-30, first data file: stack already warm at midnight, history unknown.	Kept as Idle (decided)
Start-up → Idle	1	Same day: heat-up stopped at about 215 °C and never went on.	Aborted start-up?
Start-up → Shut-down	2	2025-11-13 and 2026-01-30: heat-ups stopped before Hot standby, then cooling.	Allow as aborted start-up?
Thermal adjustment 03-13	1	After an 8-min load run ended by a trip, stack T dropped to 419 °C and climbed back at up to 207 °C/h (limit 120). Looks like a sensor or trip effect.	Check event log



Hours per stage after the fix. Rated power never occurs. Rebuild quality (R^2) unchanged.

Where it lives: data_analysis/operating_states.py (thermal_adjustment, sub_stage); flow_check.py and build_library.py (allowed arrows); operational-modes.md §1–§2. Library rebuilt: 90 episodes.

If you remember one thing: what is left is the aborted heat-ups and one suspicious re-heat on 03-13; nothing in the plant's normal running path breaks the rules any more.